RESEARCH COMPILATION 2026

The Deep Learning Canon

Perceptrons to CLIP, 1943–2017: every mechanism — convolutions, gates, residuals, replay, search — that the transformer era inherits.

Intelligence Research Atlas — Volume 02 of 6 —

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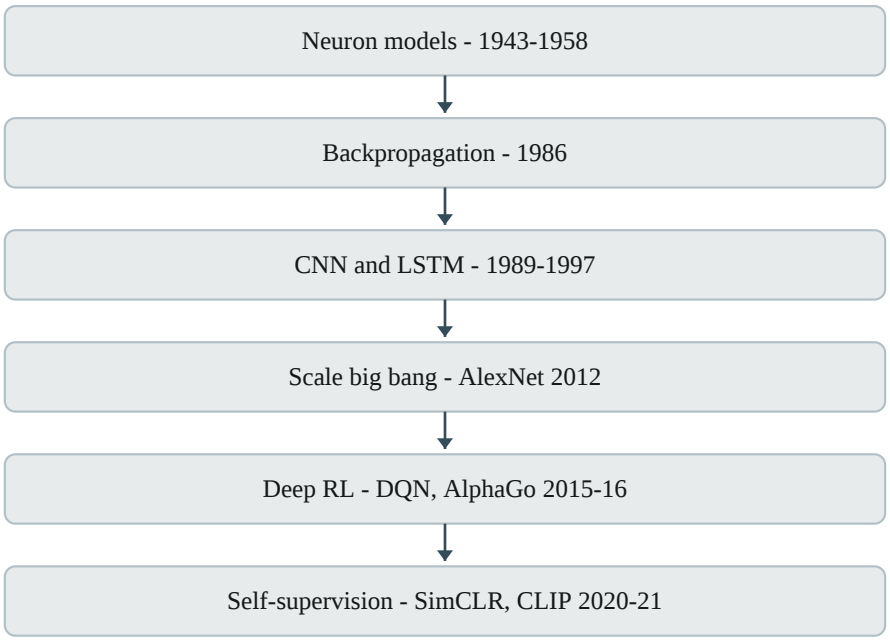
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The Deep Learning Canon

From the McCulloch-Pitts neuron to CLIP: 27 papers that built the field, and the ten findings that survived into the modern era.

The Deep Learning Canon - the through-line



Part I - Distilled Findings

F1 - The field's founding cycle: invent -> over-promise -> winter -> revive

Implication for building intelligence: Expect the current scaling era to hit its own wall (data limits, energy); the pattern says the revival will come from a mechanism currently "known but impractical" - keep a shelf of discarded-but-sound ideas (neuromorphic, symbolic hybrids, energy-based).

F2 - Architecture is crystallized inductive bias

Implication for building intelligence: Design rule for new architectures: state the world-assumption explicitly, then build the cheapest structure enforcing it (domain 01, F2 - no free lunch made rigorous).

F3 - Depth needs help: gates, normalization, and identity paths are non-negotiable

Implication for building intelligence: Checklist for any new architecture: does information have an unobstructed identity path? Are activation scales controlled? Can the system learn to ignore a layer (gate)? Miss one and depth punishes you.

F4 - Prediction-trained vectors contain semantics

Implication for building intelligence: Embeddings are the connective tissue of modern AI systems (RAG, tool retrieval, memory). Their geometry is learned, not designed - probe it empirically (domain 03 interpretability).

F5 - Self-supervision converted all data into training data

Implication for building intelligence: Labels were the bottleneck of classical ML; removing it is why foundation models exist. Any new modality (robotics, genomics, audio) should first ask: what's the self-supervised pretext?

F6 - Adversarial and minimax objectives are powerful but unstable teachers

Implication for building intelligence: Prefer minimax only where self-play genuinely explores (games, debate, red-teaming - domain 05); for representation learning, fixed-objective beats adversary.

F7 - Deep RL works when you bolt on the brain's tricks

Implication for building intelligence: When RL systems misbehave (as reasoning-model training sometimes does, domain 03), the levers are the same: buffer coverage, target stability, reward shaping. Neuroscience is a parts catalog for RL repairs.

F8 - Search atop learned evaluation beats either alone

Implication for building intelligence: This is the exact template of 2024–2026 reasoning LLMs: learned prior + test-time search (chain-of-thought as MCTS-lite). Test-time compute scaling (domain 03, F8) is AlphaZero's lesson applied to language.

F9 - Generative modeling's arc: likelihood -> adversarial -> compression -> diffusion

Evidence: VAE (Kingma & Welling 2013) introduced the ELBO; GAN (2014) sharpness without stability; diffusion (Sohl-Dickstein 2015 -> DDPM 2020) stable training with excellent samples; 2025 diffusion LLMs (domain 05) bring parallel token generation to text.

Implication for building intelligence: Choose generative machinery by data type and inference needs: autoregressive = streaming/any-time, diffusion = parallel/iterative refinement, VAE = explicit latent space. Hybrids are the frontier.

F10 - The 2012 explosion was a conjunction, not an algorithm jump

Evidence: The CNN idea existed since 1989 (LeNet); ImageNet (Fei-Fei Li's dataset, 2009) and GTX 580 GPUs (2012) were the missing legs; modern scaling-law planning (domain 01, F8) is the formalization of managing all three resources.

Implication for building intelligence: When predicting capability jumps (or designing a new system), model all three constraints: ideas are usually NOT the bottleneck - data and compute are. And post-2025, a fourth axis exists: inference-time compute (domain 03).

Part II - The Canon

27 landmark papers in chronological order. Full entries with links and keywords live in the corpus folder 02_deep-learning-canon/papers.md.

Year	Paper / Authors	Core idea	Why it matters
1949	**The Organization of Behavior** Donald Hebb	"Neurons that fire together wire together" - cell assemblies form via correlated activity.	The first learning rule; Oja's PCA (domain 01) and all Hebbian methods descend from it.
1958	**The Perceptron: A Probabilistic Model** Frank Rosenblatt	A learnable threshold neuron with a convergence proof for linearly separable data.	First artificial neural network trained by a rule; launched (and, with Minsky, froze) the field.
1969	**Perceptrons** Marvin Minsky, Seymour Papert	Proves single-layer perceptrons cannot compute XOR; (over-)read as killing neural networks.	First AI winter trigger; its real lesson - depth or hidden layers are necessary - took 17 years to answer (backprop).
1986	**Learning Internal Representations by Error Propagation (PDP, vol. 1)** David Rumelhart, Geoffrey Hinton, Ronald Williams	Multilayer networks trained end-to-end by backpropagated error signals.	Reopened the field; every gradient-trained system since (all of them) uses this.
1989	**Generalization and Network Design Strategies** Yann LeCun (constraint-based / weight sharing)	"Gradient-Based Learning Applied to Document Recognition" (LeCun, Bottou, Bengio, Haffner, Proc. IEEE) industrialized it as LeNet-5.	The convolutional prior - the strongest demonstration that architecture = inductive bias (domain 01, F2); banks read checks with this in the 1990s.
1991	**Untersuchungen zu dynamischen neuronalen Netzen (vanishing gradients)** Sepp Hochreiter (diploma thesis, advisor J. Schmidhuber)	Gradients through deep/squashing activations decay or explode exponentially - the "fundamental deep learning problem."	Named the obstacle that residuates every architecture decision for 25 years (LSTM, ReLU, residuals, normalization).
1992	**Q-Learning** Christopher Watkins, Peter Dayan	Learn action values by bootstrapping temporal-difference errors - no model of the world needed.	The bridge to neuroscience: dopamine neurons compute exactly this signal (Schultz 1997, domain 04); foundation of all value-based RL.
1997	**Long Short-Term Memory** Sepp Hochreiter, Jürgen Schmidhuber	Gated memory cells let gradients persist across hundreds of timesteps.	THE sequence model 1997–2017 (translation, speech); gates and state-carrying cells reincarnate as the transformer's residual stream with memory roles.

Year	Paper / Authors	Core idea	Why it matters
2012	**ImageNet Classification with Deep Convolutional Neural Networks** Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton	GPU-trained deep CNN cuts ImageNet error nearly in half (16.4% vs 26.2%).	The big bang of the modern era - proof that scale (data × compute × depth) wins; ignited the decade of deep learning.

Year	Paper / Authors	Core idea	Why it matters
2013	**Distributed Representations of Words and Phrases (word2vec)** Tomas Mikolov et al.	Skip-gram with negative sampling learns vector embeddings where directions carry meaning (king − man + woman ≈ queen).	Meaning as geometry - the first widely-used demonstration that prediction-trained vectors encode semantics; the "embedding" concept every LLM uses internally.
2013	**Auto-Encoding Variational Bayes** Diederik Kingma, Max Welling	Variational inference made amenable by reparameterization; encoder/decoder learn probabilistic latent representations.	Founded deep generative modeling; the ELBO objective returns in diffusion theory and in "analysis-by-synthesis" brain models (domain 04).
2014	**Sequence to Sequence Learning with Neural Networks** Ilya Sutskever, Oriol Vinyals, Quoc Le	Encode a sentence into a fixed vector, then decode it into another language.	Established the encoder-decoder paradigm - and its fixed-vector bottleneck, which attention (next entry) was invented to remove.
2014	**Neural Machine Translation by Jointly Learning to Align and Translate** Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio	Attention: the decoder dynamically looks back at all encoder states instead of one vector.	Invented attention as a soft, differentiable lookup - the mechanism the transformer made foundational (domain 03).
2014	**Generative Adversarial Networks** Ian Goodfellow et al.	Generator and discriminator locked in adversarial minimax; sample generation without explicit likelihoods.	Adversarial self-play as a learning signal - concept recycled in debate-based alignment and adversarial eval (domain 05); also the instability case study that made the field take training dynamics seriously.
2014	**Adam: A Method for Stochastic Optimization** Diederik Kingma, Jimmy Ba	Per-parameter adaptive moments; robust default step sizes.	The optimization workhorse (with AdamW variant) behind essentially all deep learning including LLM pretraining.
2014	**Neural Turing Machines** Alex Graves, Greg Wayne, Ivo Danihelka	Differentiable external memory addressed by content and location.	Opened the external-memory line (DNC 2016) - memory as an architectural component rather than emergent state; feeds directly into RAG and agent memory design (domain 05).
2015	**Batch Normalization: Accelerating Deep Network Training** Sergey Ioffe, Christian Szegedy	Normalize layer inputs per minibatch to stabilize activation statistics.	Made very deep training stable; its successor LayerNorm/RMSNorm is a fixed component of every transformer block.
2015	**Deep Residual Learning for Image Recognition** Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun	Learn residual functions with identity shortcuts -> 100+ layer networks trainable.	The residual stream is the transformer's backbone; loss-landscape geometry (domain 01, F7) explains WHY it works. Arguably the most reused architectural idea in history.

Year	Paper / Authors	Core idea	Why it matters
2015	**Human-Level Control Through Deep Reinforcement Learning** Volodymyr Mnih et al. (DeepMind)	Deep Q-network: CNN + Q-learning + experience replay + target networks learns Atari from raw pixels.	First human-level learning from raw perception; experience replay explicitly borrowed from hippocampal replay (domain 04) - the neuro-AI transfer landmark.
2015	**Deep Unsupervised Learning using Nonequilibrium Thermodynamics** Jascha Sohl-Dickstein et al.	Generate by slowly destroying structure with diffusion, then learn to reverse the noising process.	Planted diffusion; with DDPM (Ho et al. 2020, arXiv 2006.11239) became the dominant generative paradigm - and by 2025 a serious language-modeling paradigm (diffusion LLMs, domain 05).
2016	**Mastering the Game of Go with Deep Neural Networks and Tree Search (AlphaGo)** David Silver et al. (DeepMind)	Policy/value networks guided Monte-Carlo tree search; trained on human games then self-play.	Search + learned evaluation beats pure scale; the template that returns in reasoning models (o1/R1 style test-time search, domain 03).
2017	**Mastering Chess and Shogi by Self-Play (AlphaZero) / [2020] MuZero** David Silver, Julian Schrittwieser et al.	AlphaZero: self-play alone reaches superhuman Go/chess/shogi. MuZero: learns a latent world model and plans in it - no rules given.	MuZero is the proof-of-concept that LEARNED world models + planning scale - the thesis of modern world-model research (domain 05).
2017	**Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer** Noam Shazeer et al.	Route each token to a few expert FFNs - parameters grow without proportional compute.	Decoupling parameters from FLOPs; the backbone of frontier open models (Mixtral, DeepSeek-V3, domain 03).
2018	**Representation Learning with Contrastive Predictive Coding** Aaron van den Oord, Yazhe Li, Oriol Vinyals	InfoNCE: learn representations by identifying the true future among negatives - maximizes mutual information.	The single loss behind SimCLR/MoCo/CLIP and many RL representation methods; predictive coding's engineering counterpart (domain 04).
2020	**A Simple Framework for Contrastive Learning of Visual Representations (SimCLR)** Ting Chen, Simon Kornblith, Mohammad Norouzi, Geoffrey Hinton	Augment-crop-contrast with big batches and projection heads alone learns strong visual features.	Showed the recipe, not the architecture, drives self-supervised vision; killed the label bottleneck for pretraining.
2020	**Momentum Contrast for Unsupervised Visual Representation Learning (MoCo)** Kaiming He et al.	Dictionary-as-queue with momentum encoder makes contrastive learning practical without huge batches.	With SimCLR, established that contrastive pretraining transfers to detection/segmentation beyond supervised ImageNet pretraining - a capability transfer precedent for all foundation models.
2021	**Learning Transferable Visual Models From Natural Language Supervision (CLIP)** Alec Radford et al. (OpenAI)	Contrastively align images and text captions at web scale -> zero-shot classification by prompting.	Multimodal grounding; zero-shot transfer as a paradigm; the backbone of image generators' text encoders and multimodal LLMs.

Full sources with links live in the corpus: papers.md / papers.json per domain.

Where to go next

The living version of this canon lives in the corpus folder for this domain: papers.md carries every entry with full links and keywords, findings.md holds the complete distilled findings, and the diagrams folder contains Mermaid sources that render in Obsidian and GitHub. To see how this domain connects to the rest of the atlas, read the cross-domain map in 00_master-atlas; to extend the corpus, follow the conventions in AGENTS.md and re-run the index and PDF build scripts in tools/.